# FIRST TERM CONTINUOUS ASSESSMENT 1 2026/2027

| **SUBJECT:** MATHEMATICS | **SECTION(S):**A | **10TH GRADE** |
|---|---|---|

**TIME ALLOWED: 20MINUTES**

## SECTION A: OBJECTIVES

1. The total of four numbers is $1214_{five}$. What is their average expressed in base five?
   A. 303.10
   B. 242.4
   C. 141
   D. 114

2. Express the product of 0.00043 and 2000 in standard form.
   A. $8.6 \times 10^{-3}$
   B. $8.6 \times 10^{-2}$
   C. $8.6 \times 10^{-1}$
   D. $8.6 \times 10^{1}$

3. $2q3_5 = 77_8$ , find q.
   A. 0
   B. 1
   C. 2
   D. 4

4. Write 39 in base two.
   A. 100111
   B. 100101
   C. 110111
   D. 111001

5. Evaluate $(20_{three})^2 - (11_{three})^2$ in base three.
   A. 101
   B. 121
   C. 202
   D. 2020

6. Simplify $\frac{0.0324 \times 0.0064}{0.048 \times 0.12}$ and give your answer in standard form.
   A. $0.36 \times 10^{1}$
   B. $3.6 \times 10^{-2}$
   C. $3.6 \times 10^{2}$
   D. $36 \times 10^{-2}$

7. Arrange in ascending order of magnitude: $26_8$, $36_7$ and $25_9$.
   A. $25_9, 26_8$ and $36_7$
   B. $25_9, 36_7$ and $26_8$
   C. $26_8$, $25_9$ and $36_7$
   D. $36_7, 26_8$ and $25_9$

8. The sum of three numbers in base two is 11101. If the first two numbers are 1011 and 1101, find the third number.
   A. $111_2$
   B. $11_2$
   C. $1011_2$
   D. $101_2$

9. Find the product of $10110_{two}$ and $11_{two}$
   A. $1000010_{two}$
   B. $1000101_{two}$
   C. $1000110_{two}$
   D. $1001010_{two}$

10. Which of the following is the greatest?
    A. $27_{nine}$
    B. $121_{eight}$
    C. $65_{seven}$
    D. $431_{five}$

11. Multiply $12_3$ by $11_3$
    A. $202_3$
    B. $132_3$
    C. $200_3$
    D. $201_3$

12. The subtraction below is in base seven. Find ∗

**5 1 6 2**
**2 6 4 4**
**2 * 1 5**

A. 2
B. 3
C. 4
D. 5

13. In what number base is the addition

465 + 24 +225 = 1050?

A. Ten
B. Nine
C. Eight
D. Seven

14. Express the square root of 0.000144 in the standard form.

A. $1.2 \times 10^{-1}$
B. $1.2 \times 10^{-3}$
C. $1.2 \times 10^{-2}$
D. $1.2 \times 10^{-4}$

15. If $243X_5 = 366$, find x.

A. 0
B. 1
C. 2
D. 3

16. If $23_y = 1111_{two}$, find the value of y.

A. 4
B. 5
C. 6
D. 7

17. Convert ( $\frac{3}{8}$ ) ten to binary.

A. $0.011_2$
B. $11_2$
C. $0.11_2$
D. $0.111_2$

18. Express $413_7$ to base 5.

A. $2311_5$
B. $1131_5$
C. $1311_5$
D. $2132_5$

19. Change $432_5$to a number in base 3.

A. $10100_3$
B. $11100_3$
C. $11101_3$
D. $10110_3$

20. Arrange the following in ascending order of magnitude$110_2, 31_8, 42_5$.

A. $110_2, 31_8, 42_5$
B. $42_5, 110_2, 31_8$
C. $42_5, 31_8, 110_2$
D. $110_2, 42_5, 31_8$